

The output is shown in Figure 1.

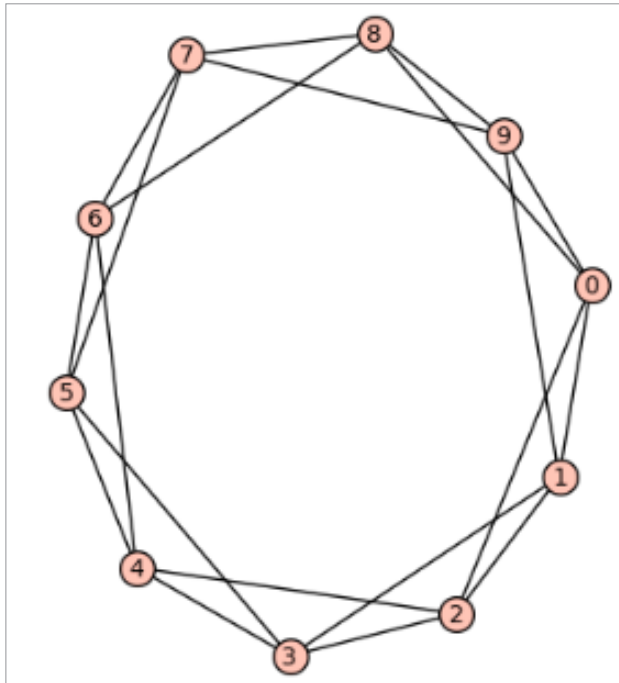


Figure 1: Vertices and edges with periodic connections

```
#Creating and visualizing a graph in SAGE
G = Graph([(1, 2), (2, 3), (3, 1)]) # Triangle graph
G.show() # Visualize the graph
#Analyzing the graph
print("Vertices:", G.vertices()) # List of vertices
print("Edges:", G.edges()) # List of edges
print("Is the graph connected?", G.is_connected())
print("Degree of vertex 1:", G.degree(1))
```

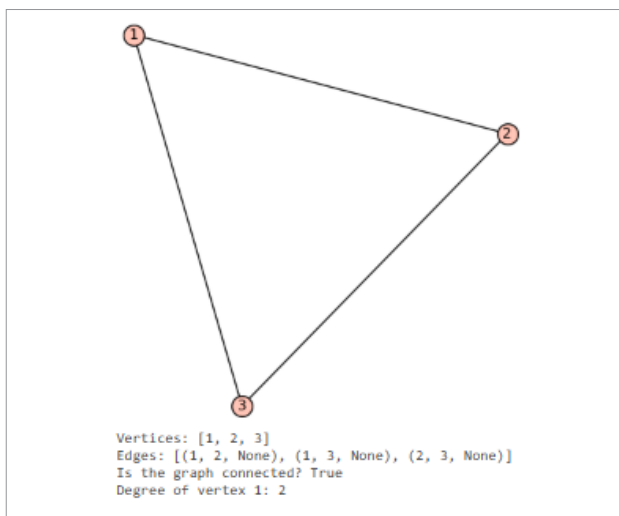


Figure 2: Output of graph analysis

```
from sage.graphs.graph import Graph
# Example Toeplitz matrix
row = [0,1,0,1,0,1,0,1,0,1]
column = [0,1,0,1,0,1,0,1,0,1]
# Get the number of vertices from the size of the matrix
num_vertices = len(row)
# Create a graph from the Toeplitz matrix with nodes labeled
from 1 to num_vertices
edges = [(i + 1, j + 1) for i in range(num_vertices) for j in
range(i + 1, num_vertices) if column[j - i] != 0]
graph = Graph(edges)
# Print the Toeplitz matrix
toeplitz_matrix = [[column[j - i] for j in range(num_
vertices)] for i in range(num_vertices)]
print("Toeplitz Matrix:")
for row in toeplitz_matrix:
    print(row)
# Draw the graph
graph.show()
# Find the domination set and its size
domination_set = graph.dominating_set()
domination_size = len(dominating_set)

print("\nDomination Set:", domination_set)
print("Domination Number:", domination_size)
```

The output is:

```
Toeplitz Matrix:
[0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
[1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
[0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
[1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
[0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
[1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
[0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
[1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
[0, 1, 0, 1, 0, 1, 0, 1, 0, 1]
[1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
```

